

Low-overhead ML methods for beamforming/combining in RIS assisted millimeter-wave massive MIMO systems

Oumaïma Bounhar

Supervised by Pr. **Ghaya Rekaya-Ben Othman**

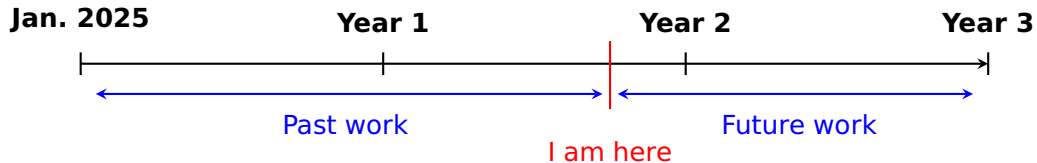
Télécom Paris, Institut-Mines-Télécom, Palaiseau, France



24 septembre 2026



Thesis Advancement & Context





Outline

Context

System Model

State-of-the-art

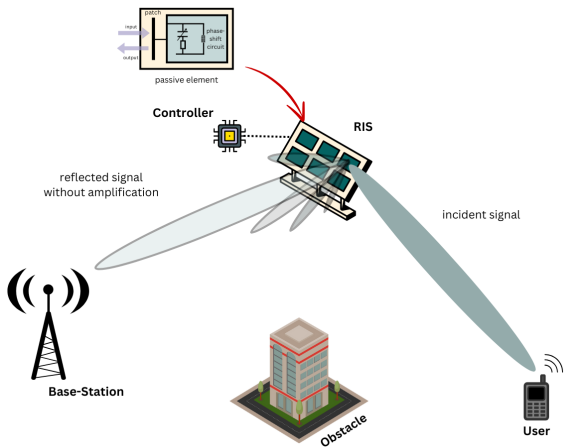
Proposed approach : Deep Reinforcement Learning

Sionna : Realistic channel generation and validation

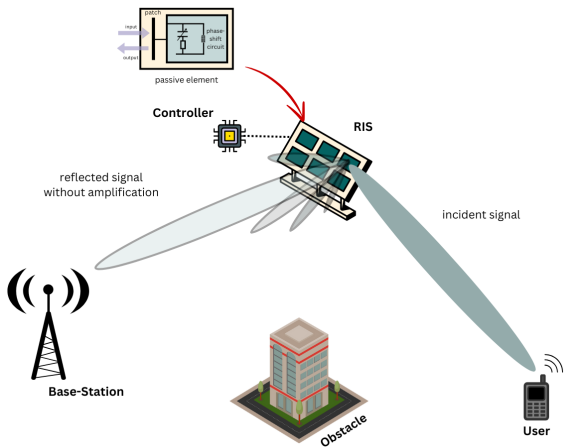
Thesis Vision Board (3 Years)

Future Perspectives

Context

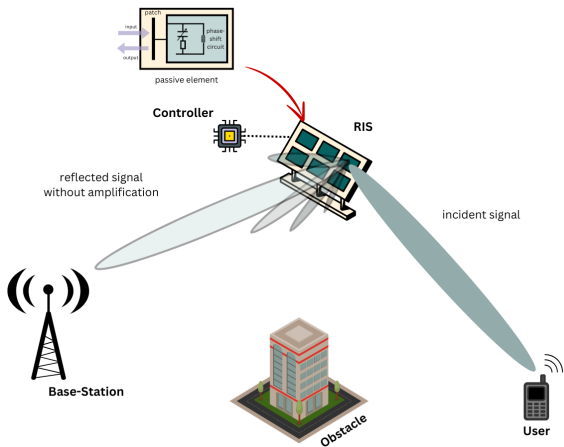


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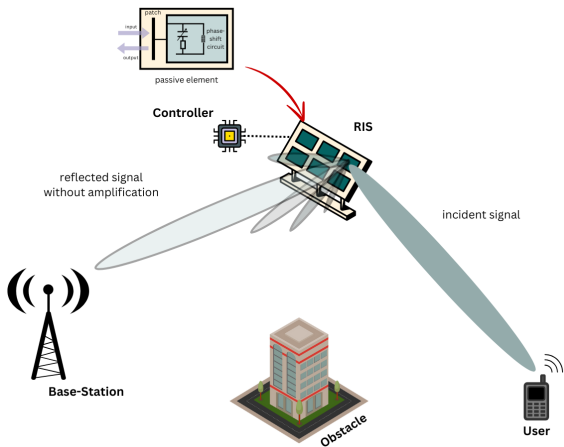
- 5G & 6G rely on millimeter waves (mmWaves) for higher coverage;

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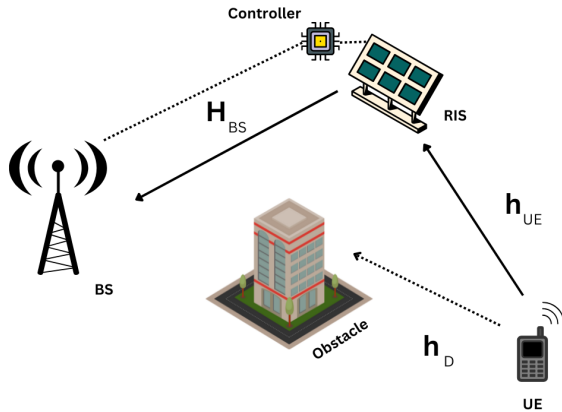
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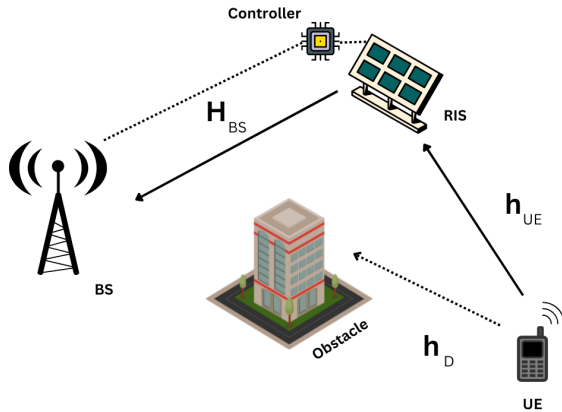
- 5G & 6G rely on millimeter waves (mmWaves) for higher coverage ;
- mmWaves suffer from fading environments ;
- RIS is a promising solution with low power consumption.

System Model



■ $N_{UE} = 1;$

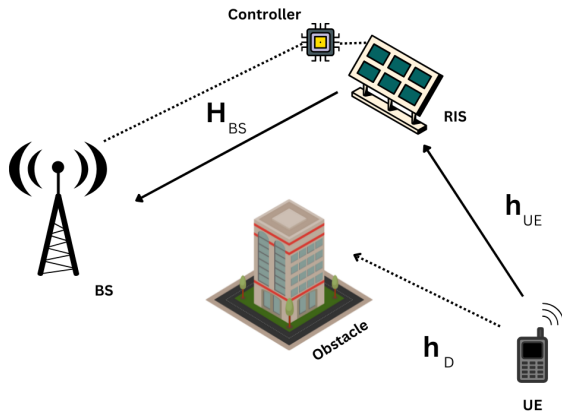
System Model



■ $N_{UE} = 1;$

■ $N_{BS} = 64;$

System Model

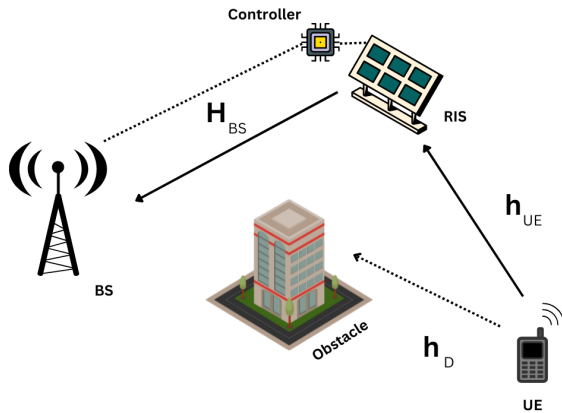


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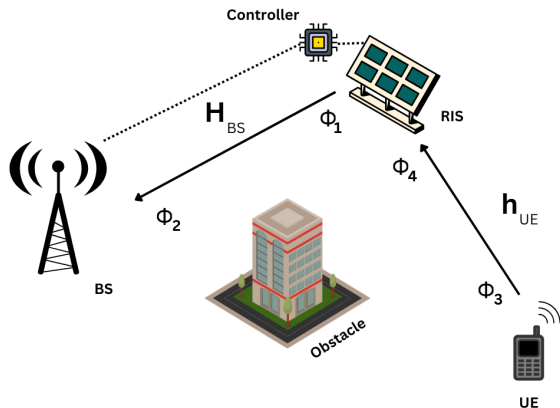
■ $N_{RIS} = 100;$

System Model



- $N_{UE} = 1$;
- $N_{BS} = 64$;
- $N_{RIS} = 100$;
- The direct link is blocked.

System Model



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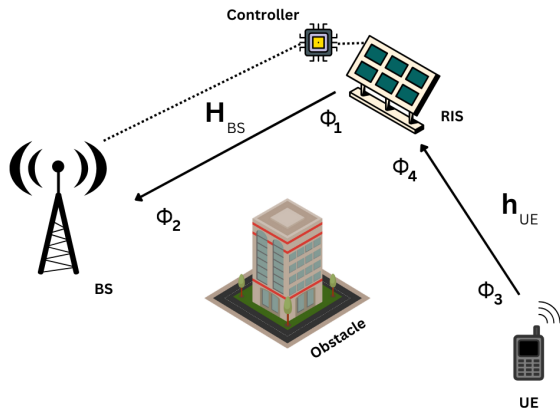
- The direct link is blocked.

We consider half-wave spaced ULAs :

$$\mathbf{H}_{BS} = \mathbf{a}_N(\phi_1)\mathbf{a}_M^H(\phi_2)$$

$$\mathbf{h}_{UE} = \mathbf{a}_1(\phi_3)\mathbf{a}_N^H(\phi_4)$$

System Model



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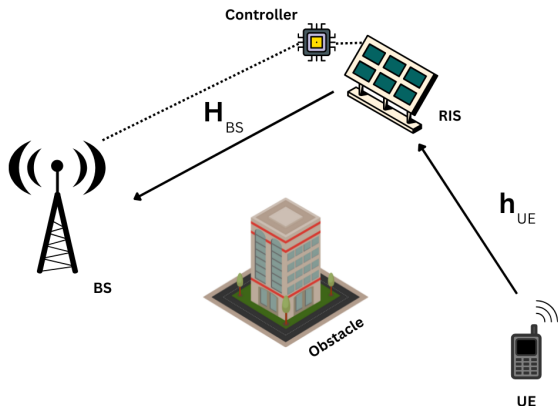
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where the steering vector is :

$$\mathbf{a}_N(\Phi) = [1, e^{j\pi \sin(\Phi)}, \dots, e^{j\pi(N-1) \sin(\Phi)}]^T$$

System Model

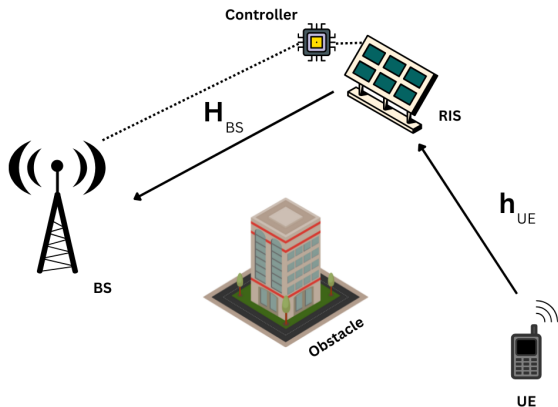


The received signal at the base station (BS) is :

$$\mathbf{y} = \sqrt{P}\mathbf{H}_{BS}\mathbf{\Theta}\mathbf{h}_{UES} + \mathbf{w}$$

where P is the transmit power of the user, $\|\mathbf{s}\|^2 = 1$, $\mathbf{w} \sim \mathcal{CN}(0, \sigma_w^2)$ and $\mathbf{\Theta} = \text{diag}(\mathbf{\Phi})$, with $\mathbf{\Phi} = [e^{j\theta_0}, \dots, e^{j\theta_{N_{RIS}-1}}]$, where the θ_i are the phase shifts introduced by the RIS, $i \in [0, N_{RIS} - 1]$.

System Model



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We write the full channel :

$$\mathbf{h}(\mathbf{\Phi}) = \mathbf{H}_{BS}\text{diag}(\mathbf{\Phi})\mathbf{h}_{UE}$$

State-of-the-art

Channel State Information (CSI) & Blind Beamforming

Definition (Rate)

We want to get a higher coverage, which leads to maximizing the rate :

$$R = \max_{i \in \llbracket N_c \rrbracket} \left[\log_2 \left(1 + \frac{P \|\mathbf{H}_{BS} \text{diag}(\boldsymbol{\Phi}_i) \mathbf{h}_{UE}\|_2^2}{\sigma_w^2} \right) \right]$$

State-of-the-art

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CSI

Channel estimation is challenging.

State-of-the-art

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CSI

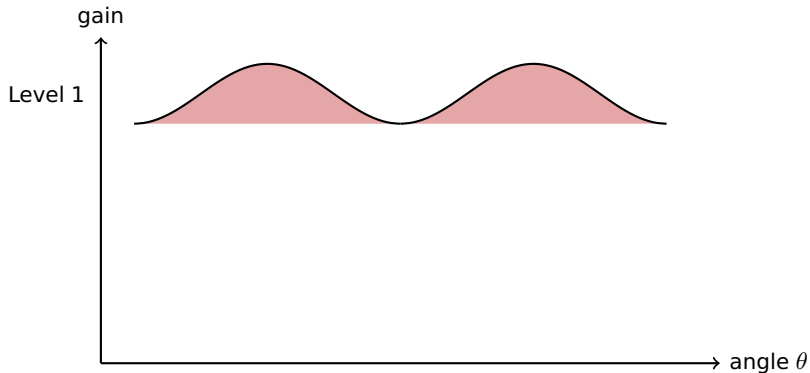
Channel estimation is challenging.

No CSI

"Blind" method.

State-of-the-art

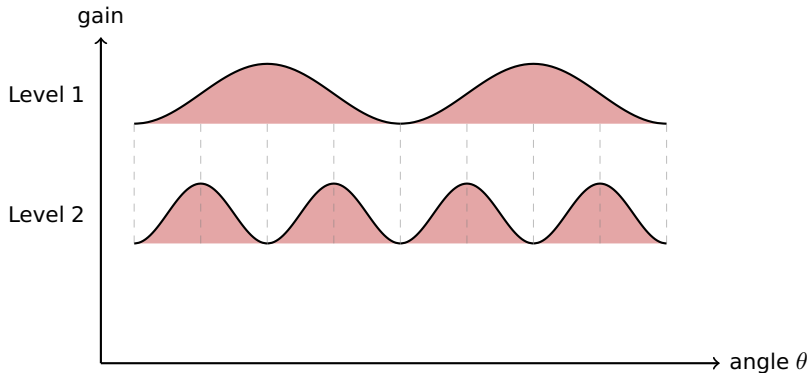
Hierarchical codebook : What Does Each Class Represent ?



Red : successive pilot search levels over the pilot codebook $\Psi_1^{N_p}$.

State-of-the-art

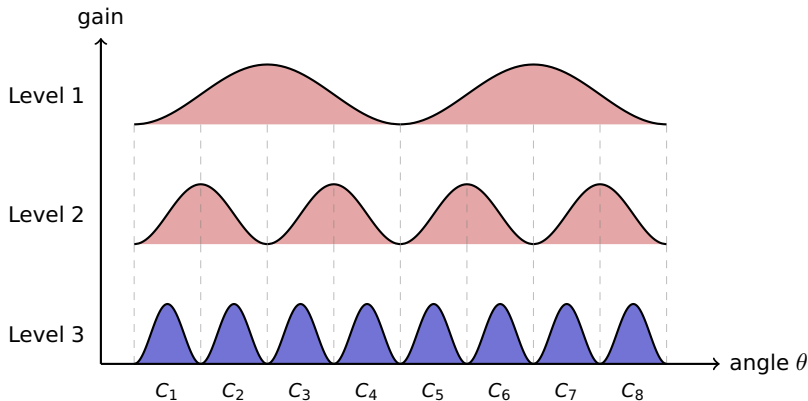
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State-of-the-art

Hierarchical codebook : What Does Each Class Represent ?



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Blue : final $N_c = 8$ narrow DFT communication beams $\rightarrow$ communication codebook $\Phi_1^{N_c} \subset \Psi_1^{N_p}$.



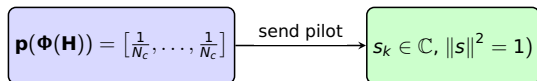
State-of-the-art

Bayesian Classifier : Protocol

$$\mathbf{p}(\Phi(\mathbf{H})) = \left[\frac{1}{N_c}, \dots, \frac{1}{N_c} \right]$$

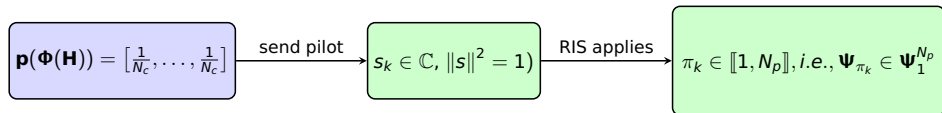
State-of-the-art

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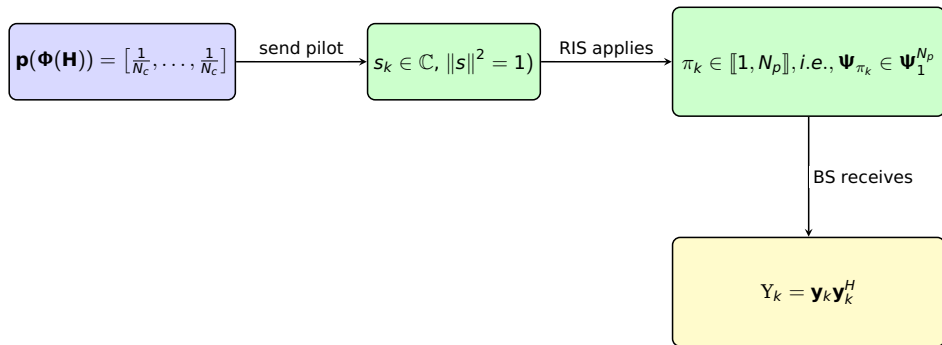
State-of-the-art

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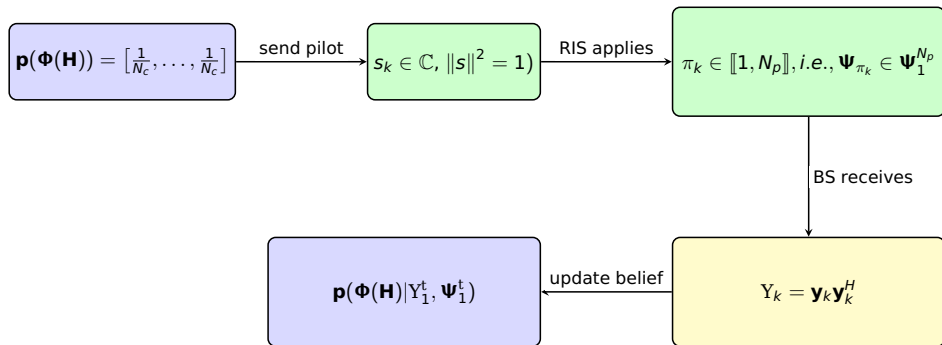
State-of-the-art

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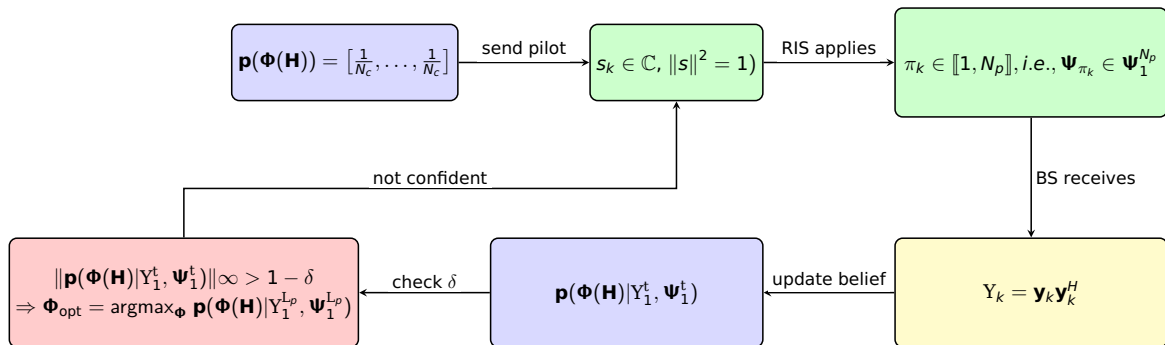
State-of-the-art

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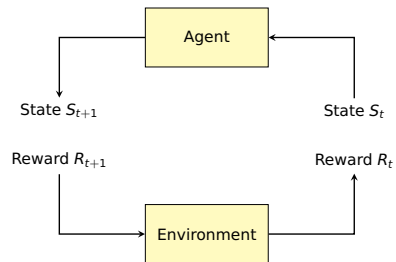
State-of-the-art

Bayesian Classifier : Protocol



State-of-the-art

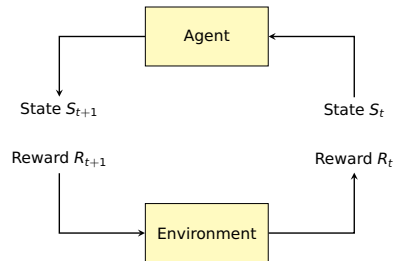
Reinforcement Learning : State, Action, and Reward Flow



State-of-the-art

Reinforcement Learning : State, Action, and Reward Flow

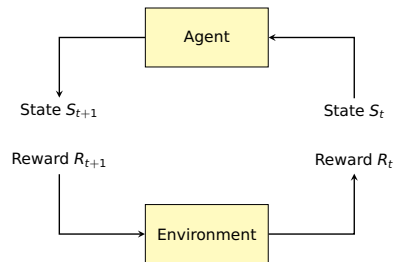
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State-of-the-art

Reinforcement Learning : State, Action, and Reward Flow

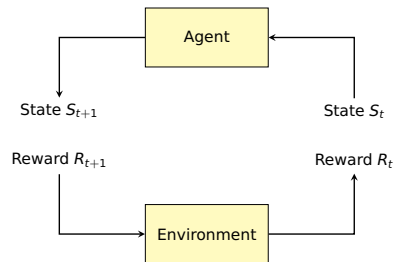
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- Action $\mathbf{a}_t \rightarrow \pi_k \in \llbracket 1, N_p \rrbracket, i.e., \Psi_{\pi_k} \in \Psi_1^{N_p}$



State-of-the-art

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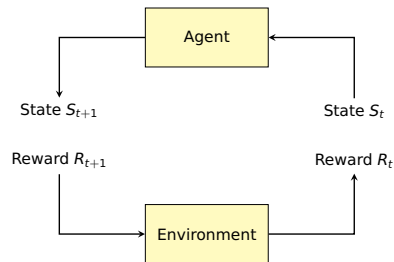
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State-of-the-art

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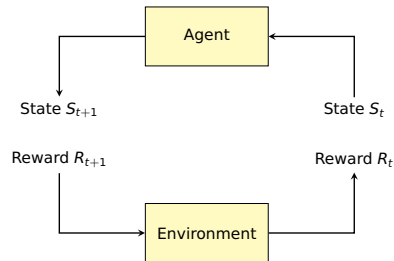
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- Reward $\mathbf{R}_t \rightarrow -1$ until reaching $\mathbf{S}_{\text{terminal}}$ where confidence level is $1 - \delta$



State-of-the-art

Reinforcement Learning : State, Action, and Reward Flow

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Objective

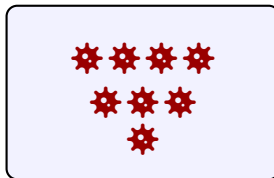
$$\Phi_{\text{opt}} = \operatorname{argmax}_{c \in \llbracket 1, N_c \rrbracket} \mathbf{p}(\Phi(\mathbf{H})|Y_1^{L_p}, \Psi_1^{L_p})$$

$$L_{\min} = \min\{L_p : \|\mathbf{p}(\Phi(\mathbf{H})|Y_1^{L_p}, \Psi_1^{L_p})\|_{\infty} > 1 - \delta\}$$

State-of-the-art

A Motivating Analogy

Reference Book (Dataset)

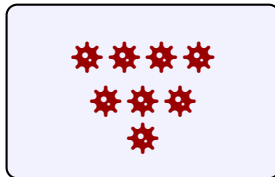


8 known diseases

State-of-the-art

A Motivating Analogy

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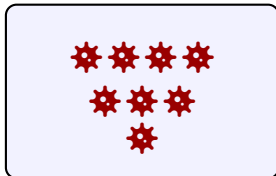


Clinician

State-of-the-art

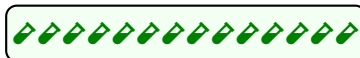
A Motivating Analogy

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8 known diseases

14 Cheap Lab Tests

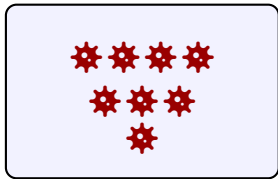


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sequentially run, one at a time

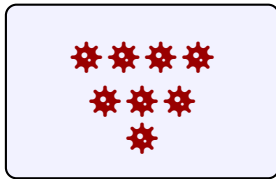


Clinician

State-of-the-art

A Motivating Analogy

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8 known diseases

compare results

14 Cheap Lab Tests



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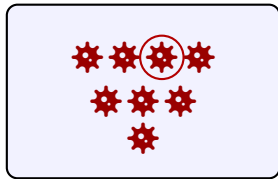


Clinician

State-of-the-art

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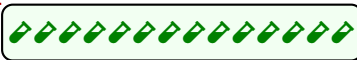


8 known diseases

Diagnosis found after
only a few tests!

compare results

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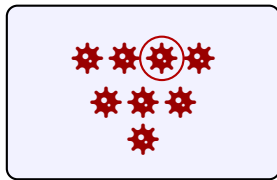


Clinician

State-of-the-art

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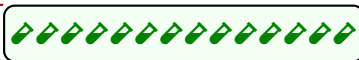


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Clinician

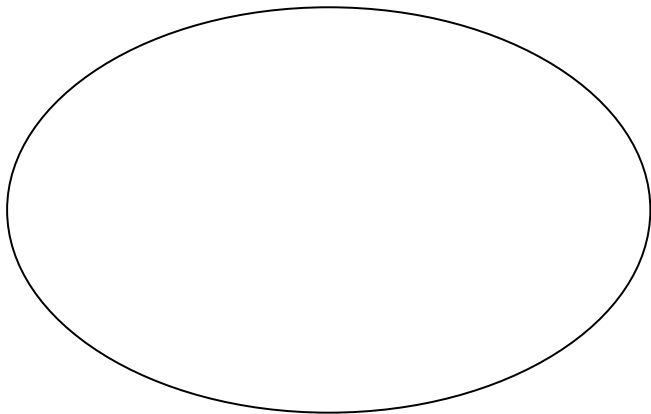
Goal : find the right diagnosis using as few cheap tests as possible.

State-of-the-art

Building the Dataset : How is a sample constructed ?

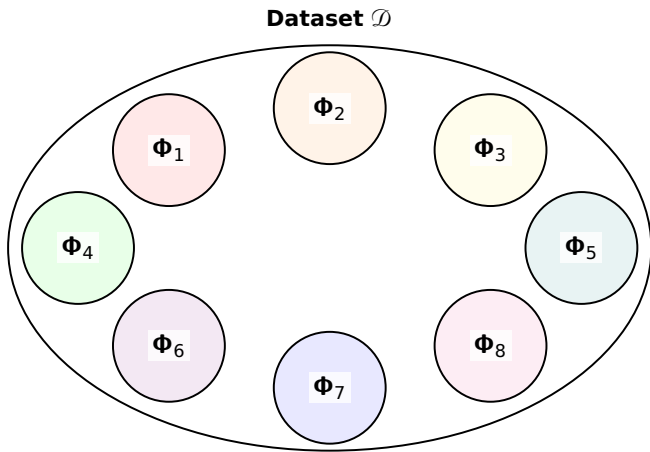
1. Draw a random channel $\mathbf{H}$.

Dataset $\mathcal{D}$



State-of-the-art

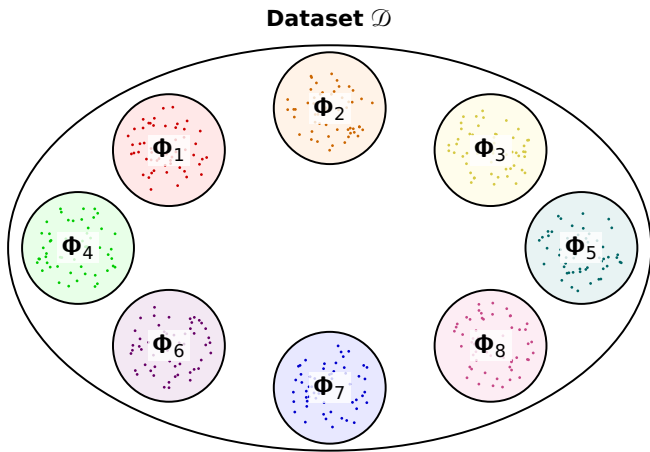
Building the Dataset : How is a sample constructed ?



1. Draw a random channel $\mathbf{H}$.
2. Run all $N_c=8$ communication codewords on $\mathbf{H}$ **without noise** (forbidden tests) $\rightarrow$ best RSE $\rightarrow$ assign c^* .

State-of-the-art

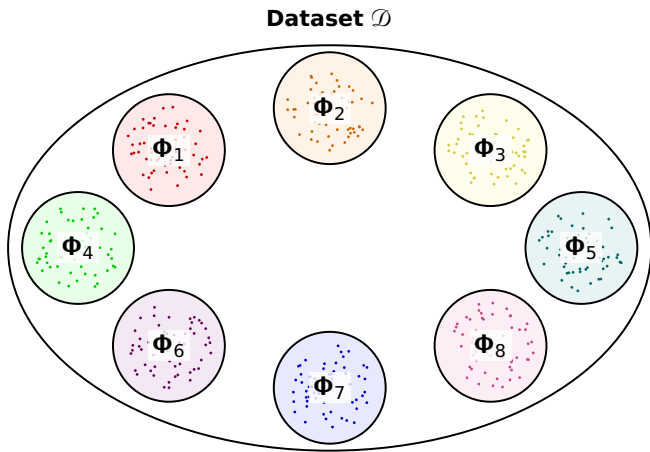
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3. Run all $N_p=14$ pilot codewords on $\mathbf{H}$ **with noise** (cheap tests) $\rightarrow$ a 14-number "fingerprint" stored as a sample of class c^* .

State-of-the-art

Building the Dataset : How is a sample constructed ?



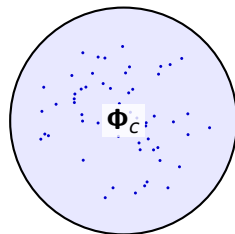
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Each class $\Phi_c \geq 200$ such samples.

State-of-the-art

The Likelihood Model

This collection **is** the likelihood model.

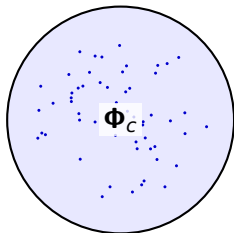


State-of-the-art

The Likelihood Model

This collection **is** the likelihood model.

“What do the 14 pilot readings
tend to look like
when the answer is c ?”



State-of-the-art

Stochastic shortest path problem : Policy, Objective & Q-Learning

Policy

A policy $\pi(S)$ determines which action A is chosen in a given state S :

$$\pi(S) = A$$

State-of-the-art

Stochastic shortest path problem : Policy, Objective & Q-Learning

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$$L_{\min} = - \sum_{t=0}^{\infty} R_t$$

State-of-the-art

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Q-Learning

Learn Q -function $Q(S, A)$ from experienced transitions (S, A, R, S') , then act greedily :

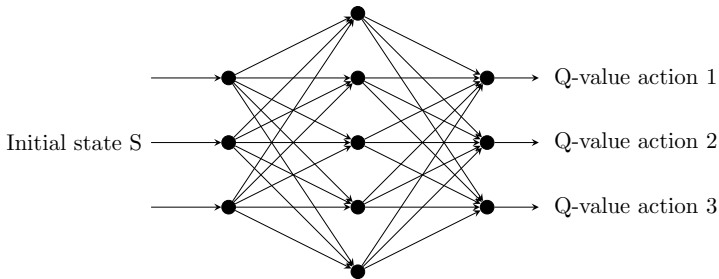
$$\underbrace{Q(S, A) \leftarrow Q(S, A) + \alpha [R + \gamma \max_{A'} Q(S', A') - Q(S, A)]}_{\text{update}} \quad \underbrace{\pi^*(S) = \arg \max_A Q^*(S, A)}_{\text{greedy policy}}$$

[1] : T. Chene, O. Bounhar, G. Rekaya, and O. Damen, "Adaptive Passive Beamforming in RIS-Aided Communications with Q-Learning," 2025.

Proposed approach : Deep Reinforcement Learning

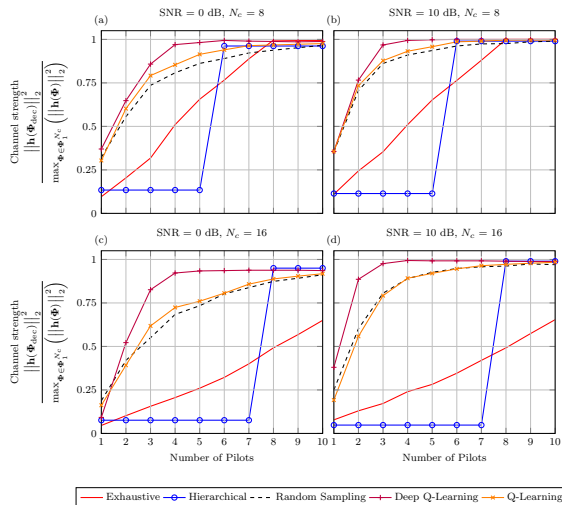
Deep Q-Neural Network

- Replacing the Q-table with a neural network (NN) that generalizes over continuous state spaces.



Proposed approach : Deep Reinforcement Learning

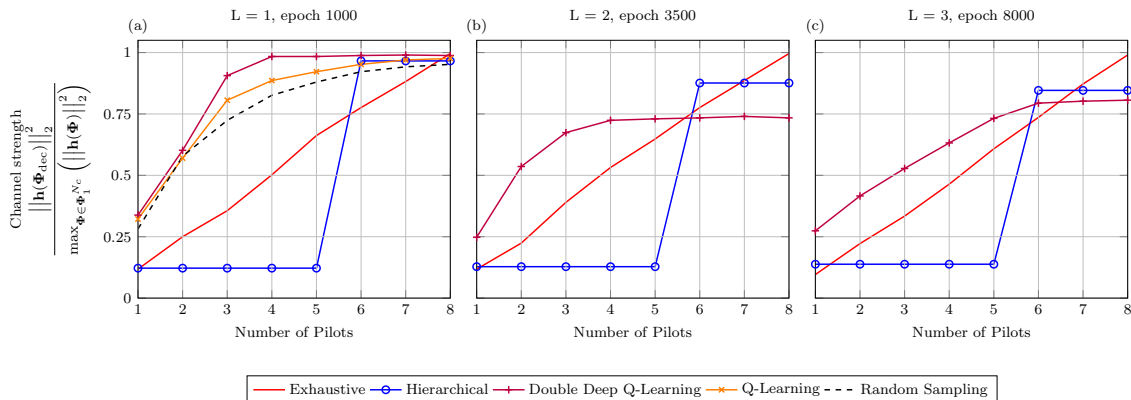
Simulation results : Comparison to other benchmarks



[2] : O. Bounhar, T. Chene, G. Rekaya, and O. Damen, "Deep Reinforcement Learning for Blind Passive Beamforming in RIS-Assisted MIMO Systems," 2026.

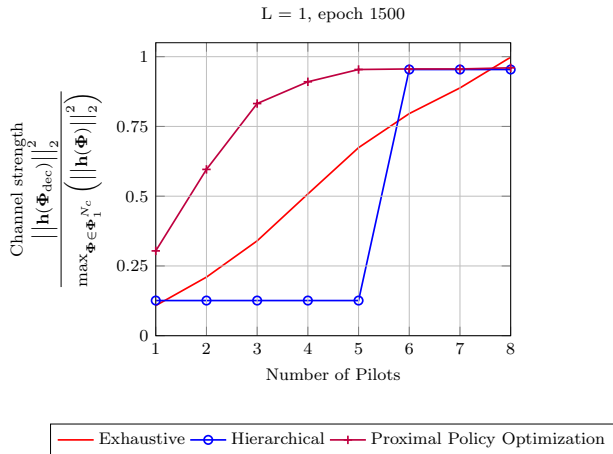
Proposed approach : Deep Reinforcement Learning

Double Deep Q-Neural Network : Comparison for SNR = 0 dB, $N_c = 8$, $\sigma_\alpha = 0$



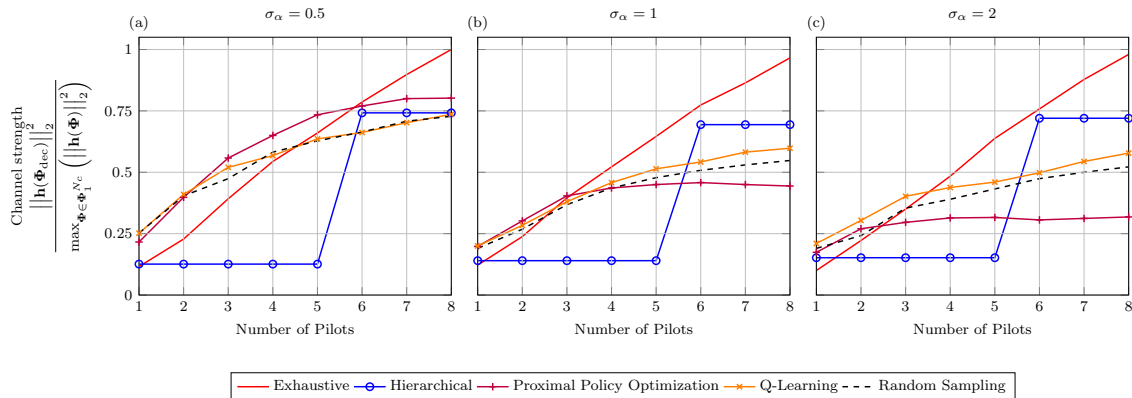
Proposed approach : Deep Reinforcement Learning

Proximal Policy Optimization : Comparison for SNR = 0 dB, $N_c = 8$, $\sigma_\alpha = 0$, $L = 1$



Proposed approach : Deep Reinforcement Learning

Proximal Policy Optimization : Sensitivity to gain variance σ_α , $L = 2$





Proposed approach : Deep Reinforcement Learning

Limitations of Deep Q-Learning

- Good for scaling size of codebook and SNR compared to QL



Proposed approach : Deep Reinforcement Learning

Limitations of Deep Q-Learning

- Good for scaling size of codebook and SNR compared to QL
- Couldn't scale for multipath -> move DDQN and PPO, they behaved better than DQN for single path but still not generalizing to multipath

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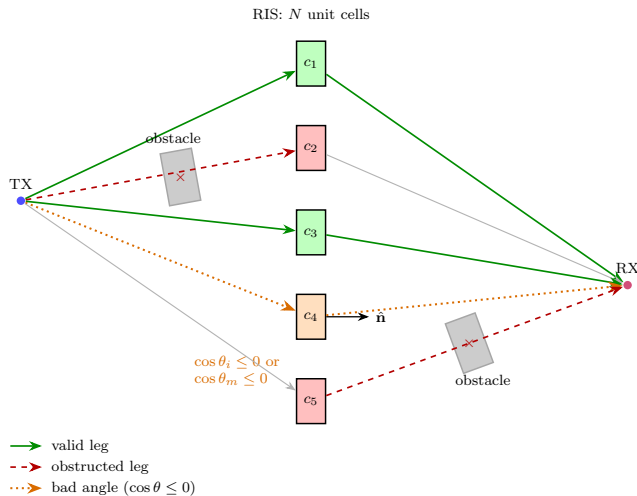
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- Need to generalize with the multipath geometric Saleh-Valenzuela channel model :

$$\mathbf{H} = \sqrt{\frac{N_t N_r}{L}} \sum_{\ell=1}^L \alpha_{\ell} \mathbf{a}_{N_r}(\phi_{\ell}^{Arr}) \mathbf{a}_{N_t}^H(\phi_{\ell}^{Dep})$$

Sionna : Realistic channel generation and validation

Ray tracing : one candidate path per unit cell



Every RIS unit cell c_n , $n = 1, \dots, N$, is treated as an **independent** candidate path :

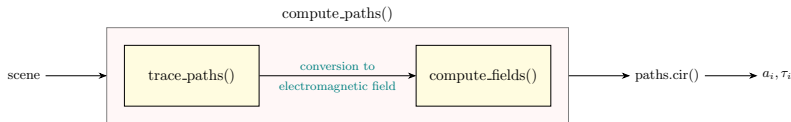
$$\text{TX} \rightarrow c_n \rightarrow \text{RX}.$$

Using the angles from the previous slide, a candidate survives only if :

- both legs are unobstructed ;
- $\cos \theta_i > 0$ and $\cos \theta_m > 0$.

Sionna : Realistic channel generation and validation

From scene geometry to channel taps



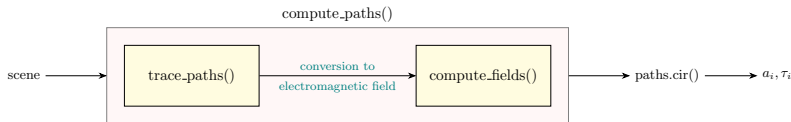
The channel impulse response is reconstructed from the N_{valid} **geometric and deterministic** per-cell taps, either kept individually resolved or coherently combined into a single tap at the mean delay :

Individually resolved (multipath)

$$h_{\text{RIS}}(t) = \sum_{n=1}^{N_{\text{valid}}} \text{coef}_n \delta(t - \tau_n)$$

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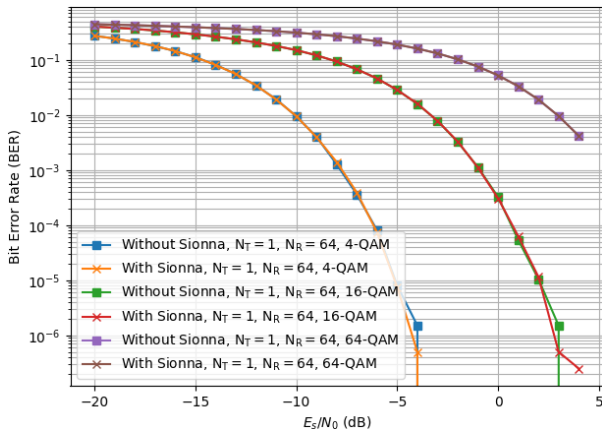
$$\bar{\tau} = \frac{1}{N_{\text{valid}}} \sum_n \tau_n$$

$$h_{\text{RIS}}(t) = a_{\text{RIS}} \delta(t - \bar{\tau})$$

$$a_{\text{RIS}} = \sum_{n=1}^{N_{\text{valid}}} \text{coef}_n e^{-j2\pi f(\tau_n - \bar{\tau})}$$

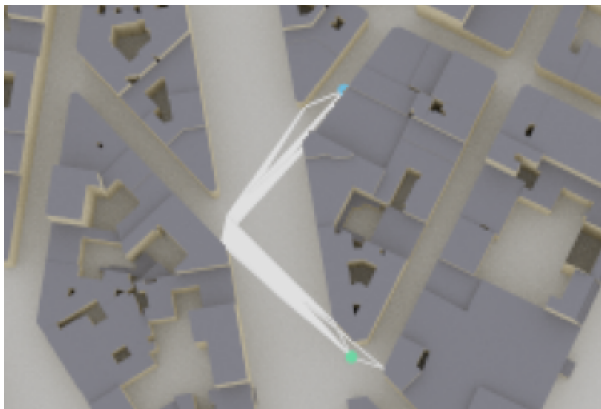
Sionna : Realistic channel generation and validation

BER validation : Sionna's Flat Fading vs. theoretical Rayleigh without Sionna



Sionna : Realistic channel generation and validation

Single-bounce per cell : RIS multipath

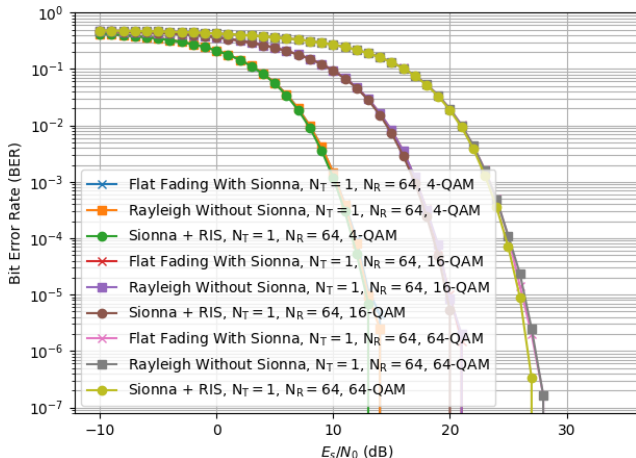


Each traced path is a single-bounce candidate $TX \rightarrow c_n \rightarrow RX$ through one RIS unit cell c_n , obtained by ray tracing (Sionna RT) over the scene. A path is kept only if it satisfies :

- **line-of-sight** on both legs ($TX-c_n$ and c_n-RX unobstructed) ;
- **positive illumination/re-radiation** : $\cos \theta_i > 0$ and $\cos \theta_m > 0$.

Sionna : Realistic channel generation and validation

BER validation : RIS multipath is correctly captured



- Baselines : theoretical Rayleigh Without Sionna and Flat Fading With Sionna overlap, as before.
- Sionna + RIS confirms the RIS's multipath is correctly captured, requiring higher SNR for the same BER, across every QAM order.

Thesis Vision Board (3 Years)

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- Problem formulation
- Q-Learning for RIS beamforming, 1st publication (WCNC 2025)
- ~~- Deep Q-Learning extension, 2nd publication (ICC 2026 ?)~~

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- Real-time Data generation (Sionna)
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- ~~Security aspects of~~ RIS ⌚
- Journal submission
- Thesis writing and defense

⚠ Focus on the choice of network that allows the protocol to scale

Future Perspectives

■ *Choice of the neural network :*

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- *Multi-user extension :* Scale the study to multi-user scenarios.

Scientific contributions



T. Chêne, O. Bounhar, G. Rekaya, and O. Damen, “Adaptive Passive Beamforming in RIS-Aided Communications with Q-Learning,” in **2025 IEEE Wireless Communications and Networking Conference (WCNC)**. Milan, Italy : IEEE, 2025, pp. 1–6.



O. Bounhar, T. Chêne, G. Rekaya, and O. Damen, “Deep reinforcement learning for blind passive beamforming in ris-assisted mimo systems,” **International Conference on Interconnected AI and NETworks (AIxNET)**, 2026.

Thank you ! Questions ?

Oumaïma Bounhar

Supervised by Pr. **Ghaya Rekaya-Ben Othman**

Télécom Paris, Institut-Mines-Télécom, Palaiseau, France



24 septembre 2026

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Goal : minimize the number of pilots L_p (pilot overhead) while reliably finding Φ_{opt} .



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What is the configuration of the RIS that maximises the rate ?

State-of-the-art

Bayesian Classifier : Protocol

For a channel realization $\mathbf{H}$, we define $\Phi(\mathbf{H})$ as the class that assigns for this channel the best communication codeword $\Phi \in \Phi_1^{N_c}$:

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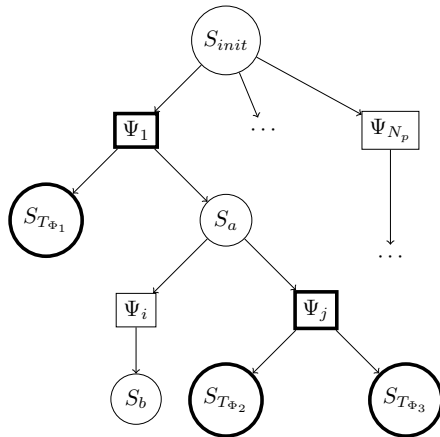
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At each time step t , update the posteriori probability with knowledge over all received feedbacks :

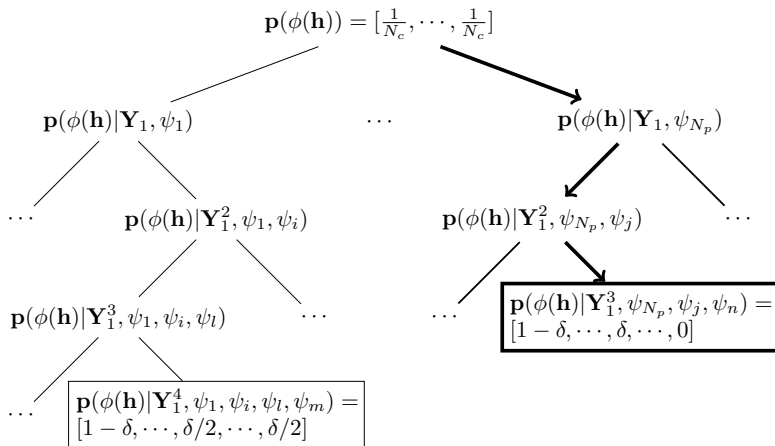
$$\mathbf{p}(\Phi(\mathbf{H})|Y_1^t, \Psi_1^t) \triangleq [\mathbf{p}(\Phi_1|Y_1^t, \Psi_1^t), \dots, \mathbf{p}(\Phi_{N_c}|Y_1^t, \Psi_1^t)]$$

Proposed approach : Reinforcement Learning

Stochastic shortest path problem



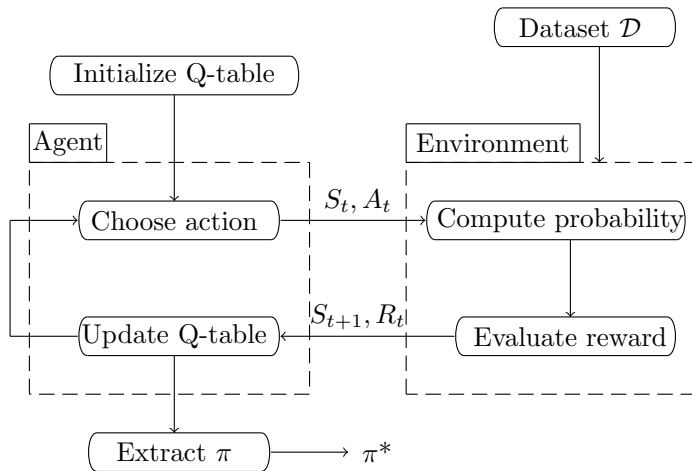
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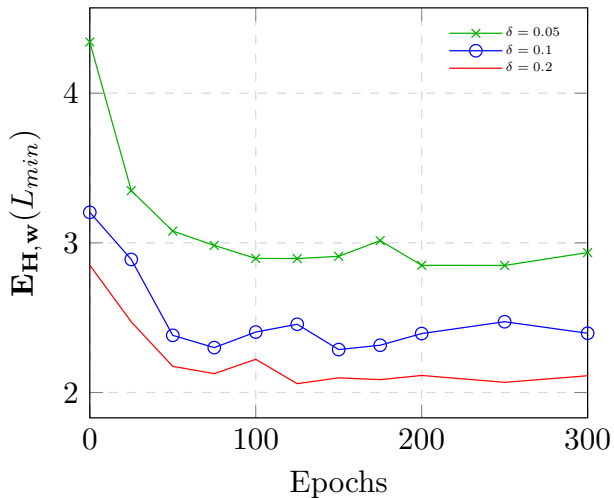
Euclidean distance

$$\mathbf{p}(\Phi(\mathbf{H}) = \Phi_k | \mathbf{Y}_1^{L_p}, \Psi_1^{L_p}) \propto \mathbf{p}(\Phi(\mathbf{H}) = \Phi_k | \mathbf{Y}_1^{L_p-1}, \Psi_1^{L_p-1}) \cdot \sum_{\mathbf{H}_1 / \mathbf{S}_T(\mathbf{H}_1)[k]=1} \frac{1}{(\mathbf{X}(\mathbf{H}_1)[L_p] - \mathbf{Y}_{L_p})^2} \quad (1)$$

Q-Learning algorithm



Stochastic shortest path problem





State-of-the-art

Blind Beamforming

Search for the optimal configuration (codeword $\Phi_{\text{opt}} \in \Phi_1^{N_c}$) of RIS over the possible configurations (communication codebook $\Phi_1^{N_c}$) :

- Exhaustive search over the codebook.
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 - Huge overhead
- Hierarchical : Need to create a specific codebook.
 - Low overhead.
 - Might not reach the optimal configuration because of its shape (binary tree).

State-of-the-art

Naive Bayes Classifier

Classification

Let L_p be the amount of the received samples, the Bayes optimal classifier is :

$$\operatorname{argmax}_{c \in \llbracket 1, N_c \rrbracket} \mathbf{p}(\Phi(\mathbf{H}) | Y_1^{L_p}, \Psi_1^{L_p})$$

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Class c_i is the class assigned to a UE direction for which communication codeword Φ_i maximizes the received signal strength $\|\mathbf{h}(\Phi_i)\|_2^2$.

State-of-the-art

Naive Bayes Classifier

Objective

Our main purpose is to find the optimal configuration Φ_{opt} while minimizing the number of required pilots $\Psi \in \Psi_1^{N_p}$:

$$L_{\min} = \min\{L_p : \|\mathbf{p}(\Phi(\mathbf{H})|Y_1^{L_p}, \Psi_1^{L_p})\|_{\infty} = 1\}$$

State-of-the-art

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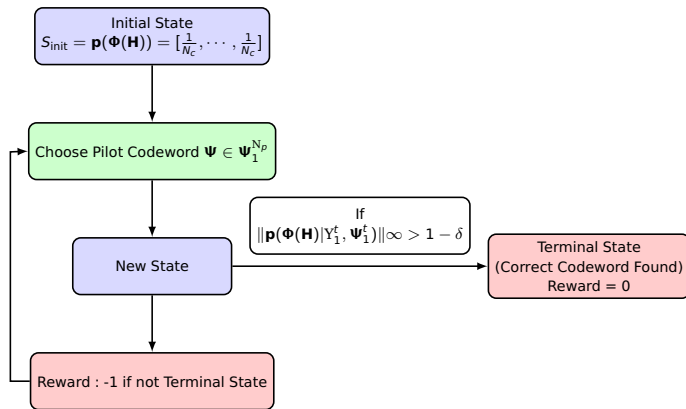
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We introduce a fixed degree of precision $\delta \in [0; 1]$ and we need to minimize L_p st. the probability of correct classification respects a certain precision :

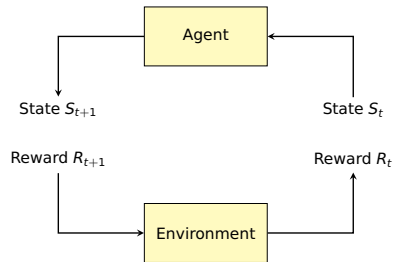
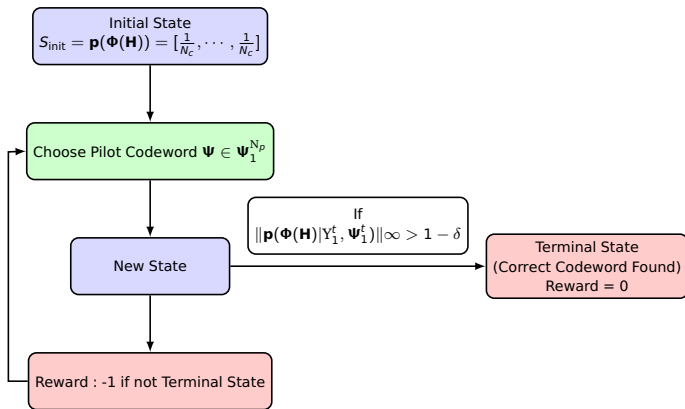
$$L_{\min} = \min\{L_p : \|\mathbf{p}(\Phi(\mathbf{H})|Y_1^{L_p}, \Psi_1^{L_p})\|_{\infty} > 1 - \delta\}$$

Proposed approach : Reinforcement Learning State, Action, and Reward Flow



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Policy

A policy $\pi(S)$ determines which action A is chosen in a given state S :

$$\pi(S) = A$$

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Objective

The goal is to reach a terminal state where the probability of correct classification respects a fixed degree of precision $\delta \in [0; 1]$:

$$L_{\min} = - \sum_{t=0}^{\infty} R_t$$

State-of-the-art

Q-Learning : Value Function and Policy

Q-function

The Q-value function represents the expected cumulative reward when taking action A_t in state S_t :

$$Q(S_t, A_t) = \mathbb{E}_{\mathbf{H}, \mathbf{w}} \left[\sum_{k=0}^{\infty} R_{t+k+1} \mid S_t = S, A_t = A \right]$$

State-of-the-art

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Bellman Equation

The Q-values are updated using the Bellman equation :

$$Q(S_t, A_t) \leftarrow (1 - \alpha)Q(S_t, A_t) + \alpha[R_{t+1} + \max_A Q(S_{t+1}, A)]$$

where $\alpha \in [0, 1]$ is the learning rate.

Optimal Policy

The optimal policy is obtained by selecting the action that maximizes the expected reward :

$$\pi^*(S) = \arg \max_A Q^*(S, A)$$

State-of-the-art

Q-Learning : Optimal policy

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The optimal policy is obtained by selecting the action that maximizes the expected reward :

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Objective

Thus, our goal is to find the optimal policy π^* that maximizes the cumulative reward despite randomness in the channel and noise. We rewrite our objective :

$$L^* = \max_{\pi} \mathbb{E}_{\mathbf{H}, \mathbf{w}} \left[\sum_{t=0}^{\infty} R_t \right]$$



Proposed approach : Deep Reinforcement Learning From Q-Learning to Deep Q-Learning

- The state space was discretized [1];

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Proposed approach : Deep Reinforcement Learning

From Q-Learning to Deep Q-Learning

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- This approach became unreliable in high dimensions;
- Small variations led to misclassifications among similar codewords.

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Proposed approach : Deep Reinforcement Learning

Deep Q-Neural Network

Deep Q-Learning (DQL) uses two NN :

- Evaluation network (Eval-Net) parameterized by the NN weights $\theta^{(eval)}$,



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- Target network (Target-Net) parameterized by NN weights $\theta^{(target)}$.

Approximation of the Q-function

The Q-function approximation can thus be expressed as :

$$Q_{eval}(S, A; \theta^{(eval)}) \approx \mathbb{E} \left[R + \gamma \max_{A'} Q_{target}(S', A'; \theta^{(target)}) \right], \quad (2)$$

Proposed approach : Deep Reinforcement Learning

Deep Q-Learning Algorithm

Epsilon-Greedy Exploration: Agent selects action: $\arg \max Q_{eval}$ with prob. $(1 - \epsilon)$, random with prob. ϵ

Experience Replay: Store transitions (S, A, R, S') in buffer $\mathcal{M}$ and sample mini-batches

Target Network Update: Periodically copy $\theta^{eval} \rightarrow \theta^{target}$ for stable learning

Proposed approach : Deep Reinforcement Learning

Deep Q-Neural Network

Network Optimization

The Q-network is trained by minimizing the mean squared error (MSE) loss, defined as :

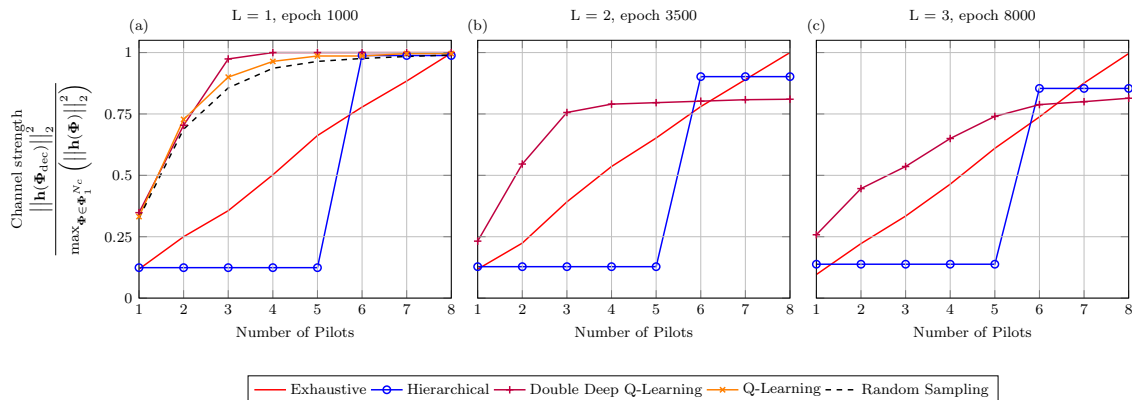
$$\mathcal{L}(\theta) = \mathbb{E}[(Q_{\text{target}} - Q_{\text{eval}}(S, A; \theta^{(\text{eval})}))^2], \quad (3)$$

with :

$$Q_{\text{target}} = R + \gamma \max_{\psi} Q_{\text{target}}(S', \psi; \theta^{(\text{target})}). \quad (4)$$

Proposed approach : Deep Reinforcement Learning

Double Deep Q-Neural Network : Comparison for SNR = 10 dB, $N_c = 8$



Proposed approach : Deep Reinforcement Learning

DQN vs. Double DQN (DDQN)

DQN

Target :

$$Q_{\text{target}} = R + \gamma \max_{\psi} Q_{\text{target}}(S', \psi; \theta^{(\text{target})})$$

The same Target-Net $\theta^{(\text{target})}$ both **selects** and **evaluates** the next action $\Rightarrow$ the max picks noisy high estimates :
overestimation bias.

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DDQN

Target :

$$Q_{\text{target}} = R + \gamma Q_{\text{target}}(S', A^*; \theta^{(\text{target})}),$$

$$A^* = \operatorname{argmax}_{\psi} Q_{\text{eval}}(S', \psi; \theta^{(\text{eval})}).$$

The Eval-Net $\theta^{(\text{eval})}$ **selects** the action, the Target-Net $\theta^{(\text{target})}$ **evaluates** it $\Rightarrow$ reduced overestimation, more stable learning.

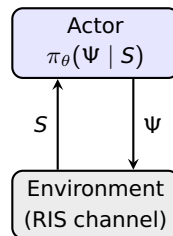
Same architecture and cost : only the target computation changes.

Proposed approach : Deep Reinforcement Learning

Proximal Policy Optimization (PPO) : Actor-Critic structure

Actor $\pi_{\theta}(\Psi | S)$

Learns the **policy** directly : a probability over the pilot codewords Ψ , given the state S .



Proposed approach : Deep Reinforcement Learning

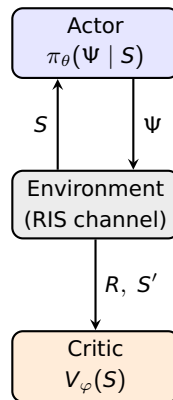
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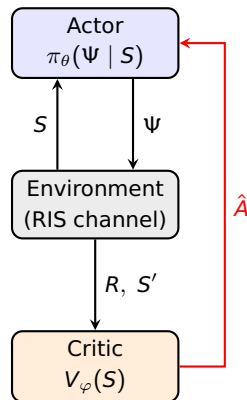
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Advantage

$$\hat{A} = R + \gamma V_{\varphi}(S') - V_{\varphi}(S)$$

$\hat{A} > 0$: Ψ did **better** than expected.



Proposed approach : Deep Reinforcement Learning

Proximal Policy Optimization (PPO) : Clipped policy update

Clipped objective

$$r(\theta) = \frac{\pi_{\theta}(\Psi | S)}{\pi_{\theta_{\text{old}}}(\Psi | S)}, \quad \mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E} \left[\min (r(\theta)\hat{A}, \text{clip}(r(\theta), 1 - \epsilon_{\text{clip}}, 1 + \epsilon_{\text{clip}})\hat{A}) \right]$$

The new policy cannot move too far from the old one in a single update $\Rightarrow$ **stable learning**.

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Training loop

1. Collect episodes with $\pi_{\theta_{\text{old}}}$;
2. Compute the advantages $\hat{A}$;
3. Update actor ($\mathcal{L}^{\text{CLIP}}$) and critic (MSE) ;
4. $\theta_{\text{old}} \leftarrow \theta$.

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vs. DQN / DDQN

- No $\max_{\Psi}$: no overestimation bias ;
- No replay buffer : learns on-policy ;
- Exploration comes from the stochastic policy, not ϵ -greedy.

State-of-the-art

The Saleh-Valenzuela channel model

To capture sparse, multipath propagation, the channel is modeled as the sum of a few paths grouped into **clusters** (macro-scatterers), each containing several **rays** (micro-scatterers) :

$$h(t) = \sum_{l=0}^L \sum_{k=0}^{K_l} \alpha_{k,l} \delta(t - T_l - \tau_{k,l})$$

- T_l : arrival time of cluster l ; $\tau_{k,l}$: arrival time of ray k relative to its cluster ;
- Clusters arrive as a Poisson process of rate Λ , rays within a cluster as a faster Poisson process of rate $\lambda \gg \Lambda$;
- Average ray power decays exponentially with both arrival times : $\mathbb{E}[|\alpha_{k,l}|^2] \propto e^{-T_l/\Gamma} e^{-\tau_{k,l}/\gamma}$;
- Each gain $\alpha_{k,l} = \beta_{k,l} e^{j\vartheta_{k,l}}$ has uniform phase and Rayleigh-distributed amplitude.

State-of-the-art

The Saleh-Valenzuela channel model

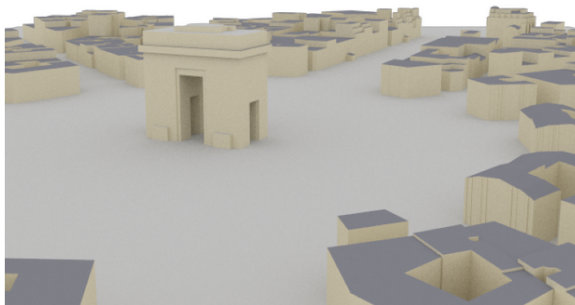
Enumerating all rays with a single path index ℓ (dropping the cluster/ray double index) and adding an **angle of departure** ϕ_{ℓ}^{Dep} and **angle of arrival** ϕ_{ℓ}^{Arr} per ray gives the geometric channel matrix used throughout this work :

$$\mathbf{H} = \sqrt{\frac{N_t N_r}{L}} \sum_{\ell=1}^L \alpha_{\ell} \mathbf{a}_{N_r}(\phi_{\ell}^{Arr}) \mathbf{a}_{N_t}^H(\phi_{\ell}^{Dep})$$

with $\mathbf{a}_N(\Phi) = \frac{1}{\sqrt{N}} [1, e^{j\pi \sin \Phi}, \dots, e^{j\pi(N-1) \sin \Phi}]^T$ the array response of a half-wavelength spaced ULA with N elements.

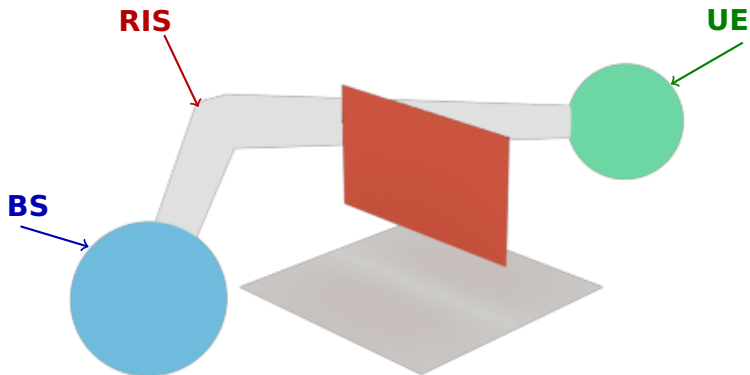
Sionna - Scene view

The area around the Arc de Triomphe in Paris



Sionna - Scene view

Simple floor-wall scenario



Visualization of the simulated scenario with a BS (64 antennas),
a RIS (1D 100 elements), and a UE (a single antenna).

Sionna - Realistic channel generation and validation

Primer on electromagnetics : RIS reradiated field

The reradiated field from the RIS at point S' is given by :

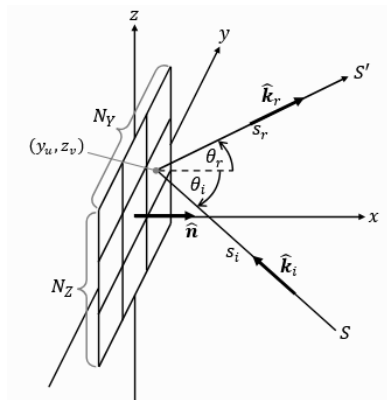
$$\mathbf{E}_r(S') = \sum_{u=1}^{N_y} \sum_{v=1}^{N_z} \Gamma_{u,v} G_{u,v} A_{u,v} \mathbf{p}_{u,v}$$

where :

$$G_{u,v} = \frac{3\lambda}{16\pi} (1 + \cos \theta_i^{u,v}) (1 + \cos \theta_r^{u,v}) ,$$

$$A_{u,v} = \frac{e^{-jk_0(s_i^{u,v} + s_r^{u,v})}}{s_i^{u,v} s_r^{u,v}} ,$$

$$\mathbf{p}_{u,v} = E_{i,\theta}(S) \hat{\theta}(\hat{\mathbf{k}}_r^{u,v}) + E_{i,\varphi}(S) \hat{\varphi}(\hat{\mathbf{k}}_r^{u,v})$$



Incident and reradiated field from a RIS

$$s_i^{u,v} = \|\mathbf{r}_{\text{RIS}}^{u,v} - \mathbf{r}_{\text{Tx}}\|_2$$

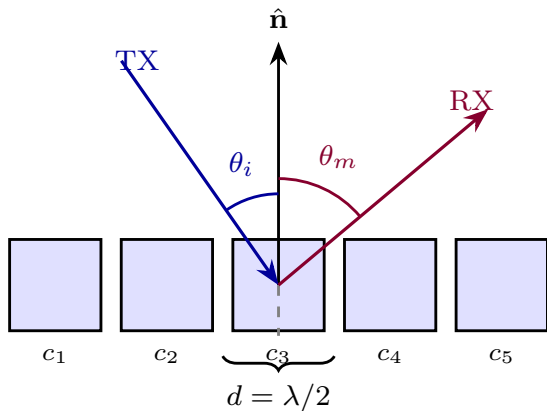
Tx-RIS distance

$$s_r^{u,v} = \|\mathbf{r}_{\text{Rx}} - \mathbf{r}_{\text{RIS}}^{u,v}\|_2$$

RIS-Rx distance

Sionna - Realistic channel generation and validation

Unit-cell geometry : incidence and reradiation angles



- Cells are spaced by half a wavelength, $d = \lambda/2$.
- Each cell sees its **own** local incidence angle θ_i and reradiation angle θ_m w.r.t. the surface normal $\hat{n}$.
- The per-cell gain scales with $(1 + \cos \theta_i)(1 + \cos \theta_m)$, and the path loss with $1/(s_i s_m)$, the Tx-cell and cell-Rx distances.

Sionna - Realistic channel generation and validation

Why geometry strongly affects the RIS channel

$$\text{Path loss} \propto \frac{1}{s_i^{u,v} s_r^{u,v}}$$

Main message

For RIS-assisted propagation, the attenuation is controlled by the both segments Tx-RIS and RIS-Rx, and it is proportional to the product of these two distances :

$$d_{\text{Tx} \rightarrow \text{RIS}} \times d_{\text{RIS} \rightarrow \text{Rx}}.$$



Sionna - Realistic channel generation and validation

Key takeaways

- In far-field regimes, per-cell delay/angle differences are tiny $\Rightarrow$ behaves like array/beamforming gain.



Sionna - Realistic channel generation and validation

Key takeaways

- In far-field regimes, per-cell delay/angle differences are tiny $\Rightarrow$ behaves like array/beamforming gain.
- In near-field regimes (large RIS, close TX/RX), delay and angular spread become physically significant $\Rightarrow$ **resolvable** multipath.